

kinquery

Challenge 8 — Find patients matching trial criteria

Which patients qualify, and why?

Team: **Jakob & Max**

The problem

a study coordinator, the afternoon a protocol lands

The PI asks: "how many of our 100 registry families would qualify?" The criteria arrive as prose:

"female, born 1965–2005, with a first- or second-degree relative with breast cancer; exclude Li-Fraumeni"

- **The answer is not in the patient record.** Family history lives in relatives and the pedigree; degree of relationship is stored nowhere.
- **Three things look like "has the disease":** a gene finding, a MIM: molecular diagnosis, an ORD0: clinical diagnosis. Trials accept some, not others.
- **"Seizures" must match *Generalized myoclonic seizure*** — an ontology question, not a string search.
- **20 of 100 probands have no date of birth.** An age criterion has to say what it did with them.

filter: id, HPO label, gene, disease, sex, year... 100 patients - 20 without DOB - 50 with gene finding - 45 molecular dx - 19 clinical dx - 31 neither gene nor dx

click a row for pedigree, relatives' records and the raw family JSON

Patient ID	Sex	DOB	Diagnosis	Gene	Relatives Count
P0003001 FAM0002001	FEMALE	1978	Breast carcinoma	BRCA1	12
P0003002 FAM0002002	FEMALE	1984	Breast carcinoma	Ovarian neoplasm	7
P0003003 FAM0002003	FEMALE	1971	Breast carcinoma	BRCA2	11
P0003004 FAM0002004	MALE	—	Breast carcinoma	Prostate cancer	10
P0003005 FAM0002005	FEMALE	—	Breast carcinoma	BRCA1	6
P0003006 FAM0002006	FEMALE	1980	Breast carcinoma	PALB2	6
P0003007 FAM0002007	FEMALE	1969	Breast carcinoma	Hereditary breast and/or ovarian cancer syndrome clinical	10
P0003008 FAM0002008	FEMALE	—	Ovarian neoplasm	Hereditary breast and/or ovarian cancer syndrome clinical	7
P0003009 FAM0002009	FEMALE	1988	Breast carcinoma	Hereditary breast and/or ovarian cancer syndrome clinical	7
P0003010 FAM0002010	FEMALE	1966	Breast carcinoma	Ovarian neoplasm	8
P0003011 FAM0002011	FEMALE	—	Breast carcinoma	—	8
P0003012 FAM0002012	FEMALE	1990	Breast carcinoma	—	8

Relatives (relationship computed from pedigree persons):

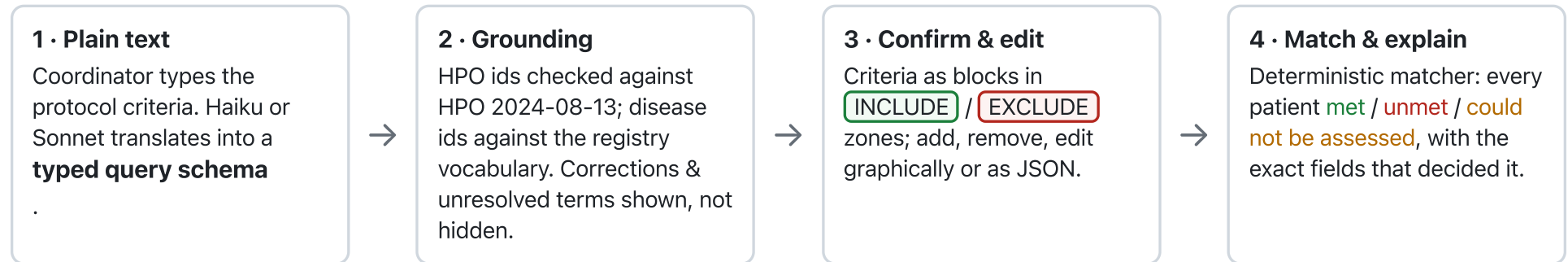
- mother (1st degree, maternal) 0 - b. 1945 - relatives[0]
- Breast carcinoma, BREAST-OVARIAN CANCER, FAMILIAL, SUSCEPTIBILITY TO, 1 [molecular]; gene BRCA1
- Hyperlipoproteinemia
- father (1st degree, paternal) 1 - b. 1945 - relatives[1]
- sister (1st degree) 9 - b. 1981 - relatives[9]
- no recorded findings
- daughter (1st degree) 10 - relatives[10]
- no recorded findings
- daughter (1st degree) 11 - b. 2011 - relatives[11]
- no recorded findings
- grandfather (2nd degree, maternal) 2 - relatives[2]
- no recorded findings
- grandmother (2nd degree, maternal) 3 - relatives[3]
- Breast carcinoma
- uncle (2nd degree, maternal) 4 - b. 1942 - relatives[4]
- Ovarian neoplasm
- uncle (2nd degree, maternal) 5 - relatives[5]
- Depression
- grandfather (2nd degree, paternal) 6 - b. 1913 - relatives[6]
- Depression
- grandmother (2nd degree, paternal) 7 - relatives[7]
- no recorded findings
- partner 9 - relatives[9]
- no recorded findings

Raw family JSON (18 KB)

The registry: 100 probands, 529 relatives, one PhenoTips family export per line.

How we addressed it

the model writes the question — code computes the answer



Query categories

- sex · birth year · age (at a reference date)
- HPO phenotype — present / explicitly excluded / absent; descendants via the ontology
- gene finding · diagnosis (molecular and/or clinical)
- family history — degree, role, lineage, count, nested conditions
- at-least-N-of · ALL · NOT; inclusion vs exclusion lists

What is computed, not stored

- **Relationship degree** from `pedigree.persons` parent links: coefficient of relationship → 1st (parent, sibling, child), 2nd (grandparent, aunt, half-sibling), 3rd (first cousin); partners never count.
- **Three-valued logic**: missing data propagates as **could not be assessed**, never as a silent exclusion.
- **Evidence**: every claim is a JSON path in the file, e.g. `relatives[3].phenotypicFeatures[0]`, plus the pedigree traversal.

The result

same query, live: 10 matched · 3 could not be assessed · 87 not matched

2 Confirm the structured query

Model's restatement (claude-sonnet-4-6; prose, not evidence — check the blocks below): Female patients born between 1965 and 2005 who have at least one first- or second-degree relative with breast cancer, excluding those with a molecular diagnosis of Li-Fraumeni syndrome.

Assumption: 'Breast cancer' in a relative is mapped to the HPO phenotype HP:0003002 (Breast carcinoma) recorded on the relative, not a disease diagnosis, since the text refers to a phenotype in a relative rather than a named syndrome.

Assumption: Li-Fraumeni exclusion applies only to molecular diagnoses (MIM:151623); clinical-only diagnoses (ORDO:524) are not excluded, as the criterion specifies 'molecular diagnosis'.

Assumption: Birth years 1965–2005 are treated as inclusive on both ends.

Assumption: Today's date (2026-09-23) is noted but age-based filtering is handled via birth_year rather than computed age.

✓ INCLUDE — a patient must satisfy ALL of these (reference date 2026-09-23)

SEX

Sex is female

✖ move to EXCLUDE

BIRTH YEAR

Born between 1965 and 2005 (inclusive) · no dateOfBirth → could not be assessed

✖ move to EXCLUDE

FAMILY HISTORY

At least 1 any sex either side relative(s), degree 1 to 2 (1 = parent/sibling/child, 2 = grandparent/aunt/uncle/half-sibling, 3 = first cousin)

► any role — click to restrict

who...

PHENOTYPE (HPO)

Patient has HP:0003002 — Breast carcinoma

include descendant terms

Matches the term and its 1 HPO descendants. Present in this registry: Breast carcinoma

Add inclusion: + Sex + Birth year + Age + Phenotype (HPO) + Gene finding + Diagnosis + Has any record + Family history + ANY / at least N of...

+ ALL of... + NOT...

and then

✖ EXCLUDE — a patient is removed if ANY of these holds; unknown → "could not be assessed"

DIAGNOSIS

Has a diagnosis with evidence molecular (MIM) clinical (ORDO) of any disease

✓ move to INCLUDE

► 1 disease term(s) selected — click to change

Structured query as INCLUDE / EXCLUDE blocks, with the model's assumptions listed for confirmation.

3 Results

10 matched

3 could not be assessed

87 not matched

100 patients

Per-criterion counts (each criterion evaluated independently over all 100 patients)

56 satisfy sex

35 satisfy birth year

16 satisfy family history

1 excluded by diagnosis

Export matched cohort (CSV)

Matched

Could not be assessed

Not matched

Patient	Sex	Born	Criteria
P0003001 FAM0002001	FEMALE	1978	sex: met · birth year: met · family history: met · exclude diagnosis: clear

Why P0003001 is met

met — sex: sex is FEMALE

self → proband.subject.sex = FEMALE

met — birth year: born 1978

self → proband.subject.dateOfBirth = 1978

met — family history: 2 of 11 relatives of degree 1-2 matched

mother (1st degree, maternal) met — Breast carcinoma: Breast carcinoma

mother (1st degree, maternal) → pedigree.persons[individualId=0] = mother (1st degree, maternal); traversal: P0003001 → maternalId → 0

mother (1st degree, maternal) → relatives[0].phenotypicFeatures[0] = HP:0003002 Breast carcinoma

grandmother (2nd degree, maternal) met — Breast carcinoma: Breast carcinoma

grandmother (2nd degree, maternal) → pedigree.persons[individualId=3] = grandmother (2nd degree, maternal); traversal: P0003001 → maternalId → 0 → maternalId → 3

grandmother (2nd degree, maternal) → relatives[3].phenotypicFeatures[0] = HP:0003002 Breast carcinoma

Exclusions (met = triggered, patient excluded):

unmet — diagnosis: no molecular diagnosis of listed diseases

Legend: highlighted (yellow), other (white), female (circle), male (square), proband (filled), dots: phenotype, pink: diagnosis, yellow: gene. Hover a person for their record.

Funnel: 56 female · 35 born 1965–2005 · 16 with a qualifying relative · 1 excluded. P0003001: mother via →maternalId→ 0 , grandmother via →maternalId→ 0 →maternalId→ 3 , both highlighted in the pedigree.

Probes: 346 HPO descendants of Seizure in scope · P0003005 "no dateOfBirth" → could not be assessed · child <10 + seizures + no molecular dx → 3 matched, 1 unassessable

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Limits, and working with Claude

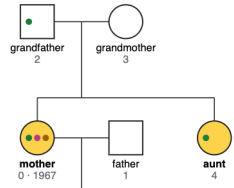
Limits (honest)

- Cohort = 100 probands. Relatives are pedigree entries, not patient records.
- Age from birth year only; most relatives have no DOB.
- No variant pathogenicity, onset age or treatments in the export — "pathogenic BRCA1 variant" becomes "BRCA1 finding", flagged as an assumption.
- Synthetic data; 100 families in memory. At scale the matcher needs an index; the pedigree maths does not change.

Patient	Sex	Born	Criteria
P0003005 FAM0002005	FEMALE	unknown	sex: met birth year: unknown family history: met exclude diagnosis: clear

Why P0003005 is unknown

- met — sex: sex is FEMALE
self → proband.subject.sex = FEMALE
- unknown — birth year: no dateOfBirth recorded
- met — family history: 2 of 6 relatives of degree 1-2 matched
 - mother (1st degree, maternal) met — Breast carcinoma: Breast carcinoma
mother (1st degree, maternal) → pedigree.persons[IndividualId=0] = mother (1st degree, maternal); traversal: P0003005 → maternalId → 0
mother (1st degree, maternal) → relatives[0].phenotypicFeatures[0] = HP:0003002 Breast carcinoma
 - aunt (2nd degree, maternal) met — Breast carcinoma: Breast carcinoma



```
graph TD
    GF[grandfather 2] --- GM[grandmother 3]
    GF --- M[mother 0 - 1967]
    GM --- M
    GM --- F[father 1]
    M --- A[aunt 4]
```

What Claude got wrong — and how we caught it

- **Hallucinated HPO ids with correct labels.** HP:0010582 "Colonic polyposis" is actually *Irregular epiphyses*; Haiku's HP:0004395 "Colonic polyps" is *Malnutrition*. Caught by ontology grounding — the label now wins; unresolvable terms are shown with candidates instead of silently used.
- **"At least two of A, B, C" approximated as OR** (12 matches instead of 3) until the schema could express it. The model itself flagged the shortcut in its assumptions list — that is how we noticed.
- **Tool output wrapped under a random key** twice; validated against the schema, never trusted.
- **Solved the wrong problem first.** Claude latched onto the pedigree — relationship maths, drawing, labels — and built a family-relationship tool. We redirected it: the user is a coordinator constructing a *cohort*; the pedigree is evidence, not the product. Inclusion/exclusion, "at least N of" and the funnel only came after that correction.
- We verified the Claude-written degree maths on a hand-drawn family (half-aunt = 3rd, first cousin = 3rd, partner = unrelated) before trusting any count.